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TVL12M

Tirunelveli District
Common Half Yearly Examination, - 2025

-12752

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Standard 12
MATHEMATICS
Part - A

Time: 3.00 Hours Marks: 90

Choose the correct answer: **20x1=20**

1) If $|adj\ (adj\ A)| = |A|^9$, then the order of the square matrix A is
a) 3 b) 4 c) 2 d) 5

2) If $A = \begin{matrix}3 & 5\\ 1 & 2\end{matrix}$, B = adj A and C = 3A then $\frac{|adj\ B|}{|c|}$ =
a) $1/3$ b) $1/9$ c) $1/4$ d) 1

3) If $a\times(b\times c)=(a\times b)\times c$ where a,b,c are any three vectors such that $b.c\neq 0$ and $a.b\neq 0$ then a and c are
a) perpendicular b) parallel
c) inclined at angle $\pi/3$ d) inclined at angle $\pi/6$

4) The co-ordinates of the point where the line $r=(6i-j-3k)+t(-i+4k)$ meets the plane $r.(i+j-k)=3$ are
a) (2, 1, 0) b) (7, −1, −7) c) (1, 2, −6) d) (5, −1, 1)

5) According to the rational root theorem, which number is not possible rational zero of $4x^7+2x^4-10x^3-5$?
a) −1 b) $5/4$ c) $4/5$ d) 5

6) The length of the diameter of the circle which touches the x-axis at the point (1, 0) and passes through the point (2, 3)
a) $6/5$ b) $5/3$ c) $10/3$ d) $3/5$

7) The locus of a point whose distance from (−2, 0) is $2/3$ times its distance from the line $x = -9/2$ is
a) a parabola b) a hyperbola c) an ellipse d) a circle

8) If $\cot^{-1}x = 2\pi/5$ for some $x\in R$, the value of $\tan^{-1}x$ is
a) $-\pi/10$ b) $\pi/5$ c) $\pi/10$ d) $-\pi/5$

9) If α and β are the roots of $x^2+x+1 = 0$ then $\alpha^{2020}+\beta^{2020}$ is
a) −2 b) −1 c) 1 d) 2

10) The order and degree of the differenential equation $\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^{1/3}+x^{1/4}=0$ are respectively
a) (2, 3) b) (3, 3) c) (2, 6) d) (4, 2)

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11) A pair of dice numbered 1, 2, 3, 4, 5, 6 of a six-sided die and 1, 2, 3, 4 of a four-sided die is rolled and the sum is determined. Let the random variable x denote this sum. Then the number of elements in the inverse image of 7 is
a) 1 b) 2 c) 3 d) 4

12) If a compound statement involves 3 simple statements, then the number of rows in the truth table is
a) 9 b) 8 c) 6 d) 3

13) The maximum slope of the tangent to the curve $y = e^x \sin x$, $x \in [0, 2\pi]$ is at
a) $x = \pi/4$ b) $x = \pi/2$ c) $x = \pi$ d) $x = 3\pi/2$

14) If $w(x, y) = x^y$, $x > 0$ then $\frac{\partial w}{\partial x}$ is equal to
a) $x^y \log x$ b) $y \log x$ c) $y.x^{y-1}$ d) $x \log y$

15) The value of $\int_{-1}^{2} |x|\, dx$ is
a) $1/2$ b) $3/2$ c) $5/2$ d) $7/2$

16) The value of $\int_{\pi/4}^{3\pi/4} \frac{dx}{1+\cos x}$ is
a) 2 b) −2 c) $1/2$ d) $-1/2$

17) Solution of differential equation $\frac{dy}{dx} = \frac{y+1}{x}$ is
a) $y = cx + 1$ b) $y = cx - 1$ c) $y = cx^2 + 1$ d) $y = cx - 2$

18) Let α and β are roots of the equation $x^2+x+1 = 0$. The equation whose roots are α^{19}, β^7 is
a) $x^2-x+1 = 0$ b) $x^2-x-1 = 0$ c) $x^2+x-1 = 0$ d) $x^2+x+1 = 0$

19) If $f(x) = \log x$ satisfies Lagrange's mean value theorem on [1, e] then the value of c is
a) $e - 3/2$ b) $e + 1/2$ c) e−1 d) $e - 1/2$

20) A die is thrown five times. Getting an odd number is considered as success. Then the variance of the distribution is
a) $8/5$ b) $3/8$ c) $4/5$ d) $5/4$

Part - B

Answer any 7 questions: Question no. 30 is compulsory: **7x2=14**

21) If $A = \begin{matrix} 8 & -4 \\ -5 & 3 \end{matrix}$, verify A(adj A) = $|A|I_2$.

22) If $z = (2+3i)(1-i)$, find z^{-1}.

23) Find a polynomial equation of minimum degree with rational co-efficients having $2+\sqrt{3}i$ as a root.

24) Find the distance of a point (2, 5, −3) from the plane $r.(6i - 3j + 2k) = 5$

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25) Verify Rolle's theorem for $f(x) = x^2 - x$, $x \in [0, 1]$

26) Assume that the shape of soap bubble is a sphere use linear approximation to approximate the increase in the surface area of a soap bubble as its radius increases from 5 cm to 5.2 cm

27) Evaluate: $\int_0^{\pi/2} \cos^7 x \, dx$

28) Find a binomial distribution for a fair die is rolled 10 times and x denotes the number of times 4 appeared.

29) Construct the truth table for $\neg p \wedge \neg q$

30) Evaluate: $\tan^{-1}\left(\sqrt{3}\right) - \cot^{-1}\left(-\sqrt{3}\right)$

Part - C

Answer any 7 questions: Question no. 40 is compulsory: **7x3=21**

31) Solve by Matrix inversion method: $5x+2y = 3$; $3x+2y = 5$

32) Find the square root of $4+3i$

33) Find the equation of tangent to the ellipse $x^2+4y^2 = 32$ when $\theta = \pi/4$

34) Find the angle between the planes $r.(i + j - 2k) = 3$ and $2x-2y+z = 2$

35) Find the intervals of monotonicites of $f(x) = 2x^3+3x^2-12x$.

36) Using Euler's theorem, if $u(x, y) = \frac{x^2 + y^2}{\sqrt{x + y}}$, prove that $x.\frac{\partial u}{\partial x} + y.\frac{\partial u}{\partial y} = \frac{3}{2}u$

37) Using integration find the volume of a right circular cone of base radius r and height h.

38) Solve: $\frac{dy}{dx} = e^{x+y} + x^3 e^y$

39) Prove that the mean and variance of a random variable x, whose probability density function. $f(x) = \begin{cases} \lambda e^{-\lambda x} & , for\ x > 0 \\ 0 & , otherwise \end{cases}$ is $1/\lambda$ and $1/\lambda^2$

40) Solve: $2x^4+9x^3+8x^2+9x+2 = 0$

Part - D

Answer all the questions: **7x5=35**

41) Solve by Gaussian elimination method

$2x-2y+3z = 2$; $x+2y-z = 3$; $3x-y+2z = 1$

(OR)

Find the area of the region bounded by $y = \cos x$, $y = \sin x$, the lines $x = \pi/4$ and $x = 5\pi/4$

42) Prove by vector method $\sin(\alpha+\beta) = \sin\alpha \cos\beta + \cos\alpha \sin\beta$

(OR)

Expand $\log(1+x)$ as a Maclaurin's series upto 4 non-zero terms for $-1 < x \leq 1$

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43) Solve the equation $x^3-9x^2+14x+24 = 0$ if it is given that two of its roots are in the ratio 3:2.

(OR)

Solve: $(x^2-3y^2)dx+2xy\,dy = 0$

44) Prove that $\tan^{-1}x+\tan^{-1}y+\tan^{-1}(z) = \tan^{-1}\dfrac{x+y+z-xyz}{1-xy-yz-zx}$

(OR)

If $w(x, y) = xy+\sin(xy)$, then prove that $\dfrac{\partial^2 w}{\partial y \partial x} = \dfrac{\partial^2 w}{\partial x \partial y}$

45) Find the equation of the circle passing through the points (1, 1) (2, −1) and (3, 2)

(OR)

A random variable x has the probability mass function

x	1	2	3	4	5
f(x)	k^2	$2k^2$	$3k^2$	2k	3k

Find (i) the value of k (ii) $P[2 \le x < 5]$ (iii) $P[3 < x]$

46) If $z = x+iy$ is a complex number such that $\operatorname{Im}\dfrac{2z+1}{iz+1} = 0$, show that the locus of z is $2x^2+2y^2+x-2y = 0$

(OR)

Verify (i) closure (ii) commutative (iii) associative (iv) identity (v) inverse for the operation X_{11} on a subset $A = \{1, 3, 4, 5, 9\}$ of the set of remainders $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

47) Find the shortest distance between the lines

$r = i + j + \lambda(2i - j + k)$ and $r = 2i + j - k\,\mu(3i - 5j + 2k)$

(OR)

Find the probability distribution of number of doublets in three throws of a pair of dice.

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